

but to core intellectual property, AI-driven chip development workflows, and talent creation across the semiconductor ecosystem.

In this exclusive conversation for our semiconductor cover story, Jain cuts through the noise around fabs and subsidies to explain what India is genuinely building. Edited excerpts of my interview follows:

What's happening on the ground in India's semiconductor industry?

Alok: We develop software to help design electronic chips. My team spans India, Israel, and the US.

Cadence India has multiple sites: Noida is the largest (about 2,000–2,500 employees) and is a microcosm of Cadence. Some technologies are corporate owned with India contributing, some are jointly owned, and some are fully owned from India.

On the National Semiconductor Mission, we have partnered with AICTE to roll out a new VLSI curriculum to 120+ universities; under Chip-to-Startup (C2S), provided a bundle of 100 Cadence tools to startups/universities with a plan to train ~85,000 engineers by 2027–28; under DLI, Cadence tools are available via the National EDA Grid to participating startups.

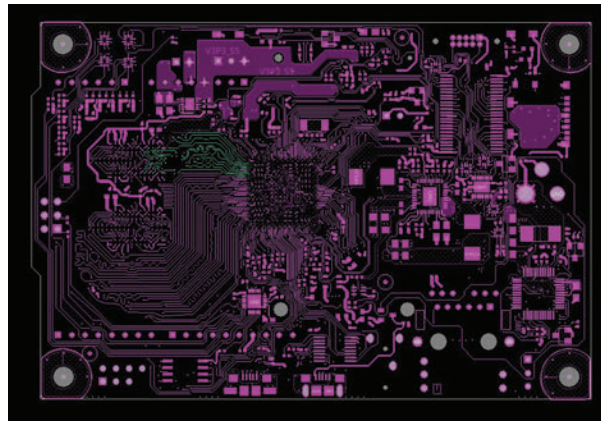
We are engaged in major universities' talent pipeline programs (hire from campuses and train into full-time roles); partnerships with IIT Delhi, IIIT Delhi, PEC, IIT Hyderabad, etc., for training, talks, research, and hiring. On the startups front, we are partnering with incubators like FabC (IIT Hyderabad) and Electro Park (Delhi and Bhubaneswar); Cadence-LIVE events for networking and best-practice sharing.

What clarity are you providing to the Indian government with respect to semiconductors?

Alok: We fully support the National Semiconductor Mission. Government focus areas – manufacturing,

assembly, and manpower development – are important. However, manufacturing and assembly are very capital-intensive, and catching up with advanced foundries (TSMC, Intel Foundry, Samsung Foundry, Global-Foundries) is a long journey.

We recommend a stronger emphasis on software/EDA development in India. It's less capital-intensive, aligns with India's strengths, and with the AI revolution we're at a tipping point where India can lead. This mainly requires trained engineers, which India has. The government is listening, but it takes time to implement.



Given India's software strength, how will we become a hardware manufacturing success?

Alok: India is moving toward hardware design, particularly at older geometry nodes and for application-specific chips – RISC-V is helping. Pace should increase, but progress is happening. Moving to lower geometry nodes and complex chips will take time.

Assembly and manufacturing at older nodes should happen soon (2–5 years). Competing at lower geometry nodes with leading foundries will take a long time and significant investment. We can learn from the US, Taiwan, Singapore, China, and leapfrog some steps, but not all – this is hard physics. We can't jump straight to 1.8-nm. We must go through the learning phases.

What do people not grasp about chip design/manufacture and the semiconductor industry?

Alok: Chip design is very complex and requires a global ecosystem. No single country – even the US – is self-sufficient. You need foundries like TSMC for manufacturing, EDA vendors, and lithography (ASML is the sole provider of the most advanced machines with millions of parts).

India must take a holistic approach – advance software, chip design, IP, manufacturing, and assembly together. Focusing on just one or two pillars won't solve it. Public

and government understanding is improving (eg, awareness of ASML's critical role).

Can you juxtapose your early career with what fresh graduates use for IC design today?

Alok: Chip design progressed from layouts to transistors to gates to RTL, I worked mostly around RTL and am aware of paper-based layout days. Today, engineers use AI and LLM tools – much of the design involves prompting in English (not fully automated, but significant).

Personally, I was a classical C/C++ engineer. Now some engineers aren't even C/C++ savvy – they specify in English, craft the right LLM prompt, and get the code or artifact they need. Things have changed substantially (I used punch cards during my masters).

Skills we now emphasize: Core CS/EE fundamentals (data structures, algorithms, CMOS, etc.), Understanding of EDA tools (verification, analog, placement, etc.), New chip/system concepts (e.g., 3DIC), AI/ML/LLMs – now mandatory to avoid becoming "antiquated." 